

Agentic AI-Based Dynamic Tariff Optimisation for EV Charging Networks

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The Problem with Flat-Rate Pricing

Static tariffs are blind to real-world dynamics. Charging stations on fixed ¥/kWh pricing face

- peak-hour congestion and user wait times
- charger under-utilisation during off-peak windows
- rising electricity procurement costs
- no incentive for demand smoothing

This project builds a three-agent agentic framework that autonomously predicts demand, sets optimal tariffs, and learns from outcomes.

Two Datasets, Two Scales

ACN-Data (Caltech / JPL, USA)

- 16,304 workplace EV charging sessions, Apr – Dec 2018
- Session-level records with timestamps, kWh delivered, station IDs
- Used for session-level EDA and per-user premium modelling

ST-EVCDP / UrbanEV (Shenzhen, China)

- 247 stations $\times$ 8,640 timesteps at 5-minute intervals
- Matrices for occupancy, volume (kWh), price, duration, adjacency, and inter-station distances
- Core dataset for demand modelling, tariff optimisation, and network analysis

Preprocessing Decisions

ACN

- Timestamps parsed from GMT and localised to US/Pacific
- Sessions with zero or negative duration filtered out
- Missing kWhRequested filled with per-station median
- Derived fields: session duration, idle duration, average power draw

ST-EVCDP

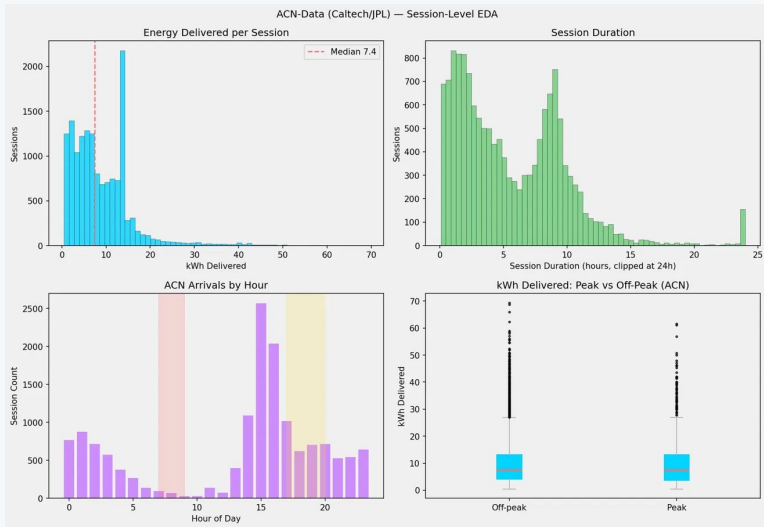
- Four wide matrices joined on (datetime, station_id)
- Station metadata merged from `information.csv`
- Missing panel values filled with station-level temporal median
- All lag and rolling features computed after sorting by station and time to prevent future leakage

Engineered Features at a Glance

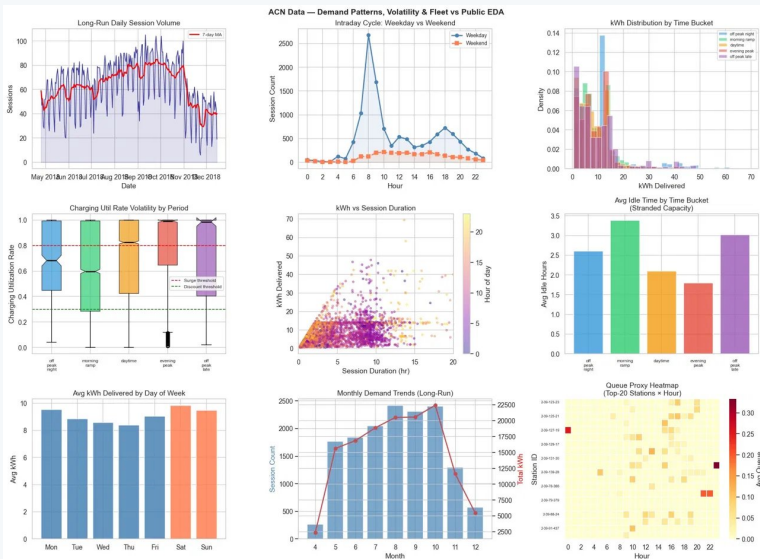
Every feature has an economic or operational grounding.

- **Utilisation rate** $u_{s,t} = \text{busy piles} / \text{capacity}$
- **Queue proxy** $\max(\text{busy piles} - 0.9 \times K_s, 0)$
- **Cyclic time encoding** $h_{\sin} = \sin(2\pi h/24), h_{\cos} = \cos(2\pi h/24)$
- **Lag features** at 5-min, 1-hour, and 24-hour horizons
- **Rolling mean and volatility** over 1-hour windows
- **Spatial spillover** $\text{Spill}_{s,t} = \sum_{j \in \mathcal{N}(s)} W_{sj} u_{j,t-1}$
- **Station efficiency index** composite score in $[0, 1]$

ACN Session-Level EDA



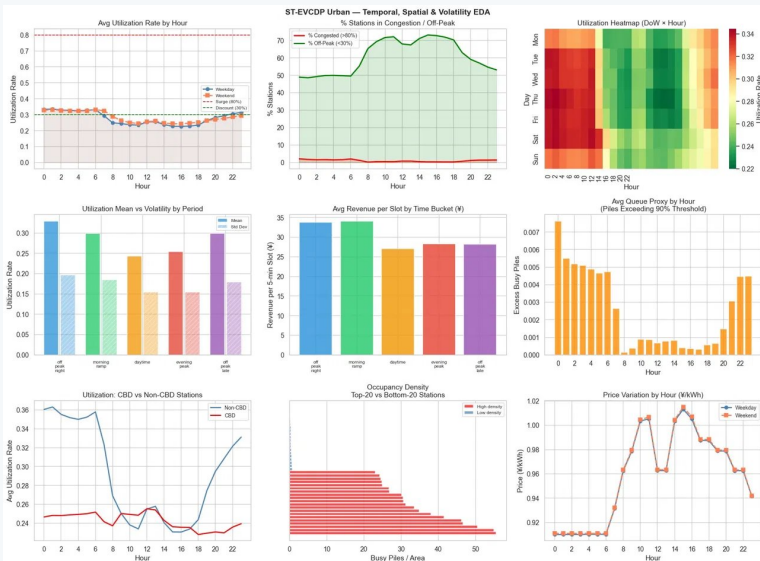
ACN Demand Patterns and Volatility



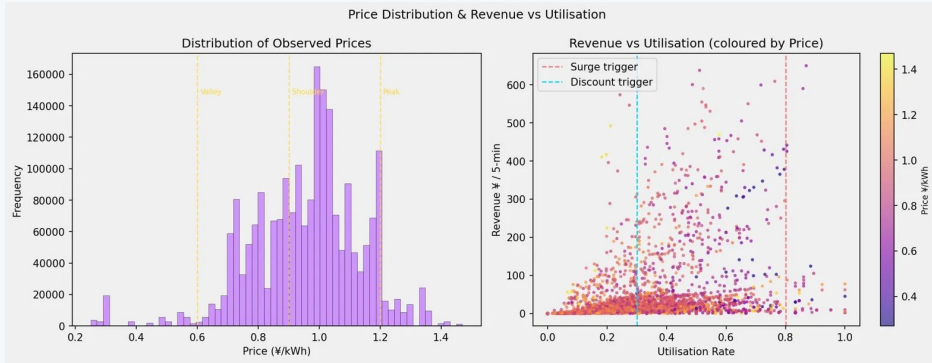
Key ACN Findings

- Strong bimodal arrival pattern: morning plug-in (8–9 am) and afternoon return (3–5 pm), driven by the workplace fleet context
- Median energy delivered is 7.4 kWh; the distribution is right-skewed with a long tail from fleet vehicles topping up overnight
- Session duration peaks between 5 and 10 hours, confirming that most users leave vehicles plugged in far beyond active charging time
- Idle time (session duration minus charging time) averages 2–3.5 hours depending on time bucket, representing stranded capacity
- Queue proxy is concentrated at stations 2-39-127-19 and 2-39-130-28 during the 14–18 hour window, pointing to two hotspot stations for surge pricing intervention

Urban (ST-EVCDP) Temporal and Spatial EDA



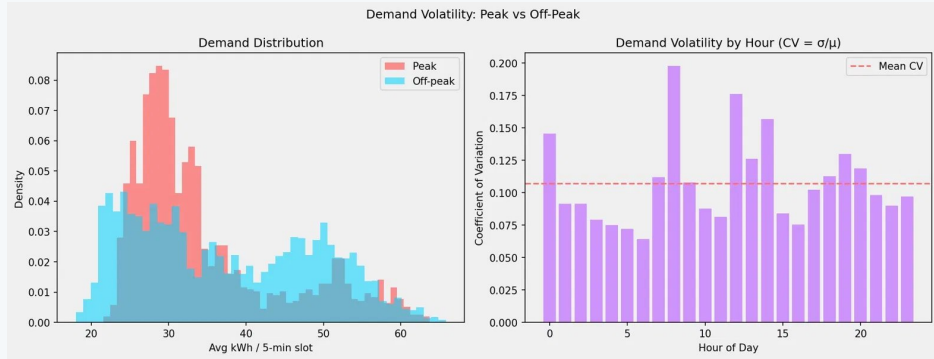
Urban (ST-EVCDP) Price Distribution and Revenue



Key Urban Demand Findings

- Utilisation across all 247 stations averages around 0.25–0.33, well below the 80% surge threshold, indicating systemic under-utilisation
- Non-CBD stations show a sharp morning dip around 10 am and a late-night spike after 22:00, while CBD stations remain flatter throughout the day
- Over 65% of station-slots are in the off-peak regime at any given hour, confirming large headroom for discount-driven demand uplift
- Price already varies from ¥0.3 to ¥1.45 per kWh in the observed data, suggesting some form of time-of-use pricing is in practice – but not systematically optimised
- Demand volatility (CV) peaks at 7–8 am and 12–13 pm, exactly the transition periods where mis-pricing is most costly

Demand Volatility: Peak vs Off-Peak



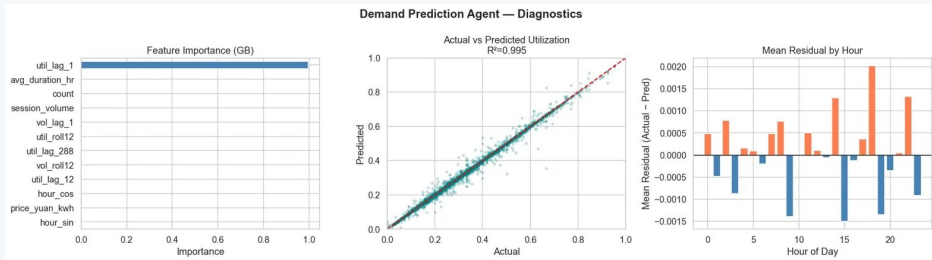
Three Models for Demand Prediction

All models target **utilisation rate** $u_{s,t} \in [0, 1]$ with a chronological 80/20 train-test split.

- **Gradient Boosting Regressor** sequential ensemble of shallow trees, each fitting the residual of the previous step
- **Random Forest Regressor** independent ensemble of deep trees on bootstrapped samples and random feature subsets
- **ST-GNN** graph convolutional encoder capturing spatial dependencies combined with a GRU for temporal dynamics

The train set contains the first 80% of timesteps; the test set contains the most recent 20%, preserving temporal ordering throughout.

Gradient Boosting Results

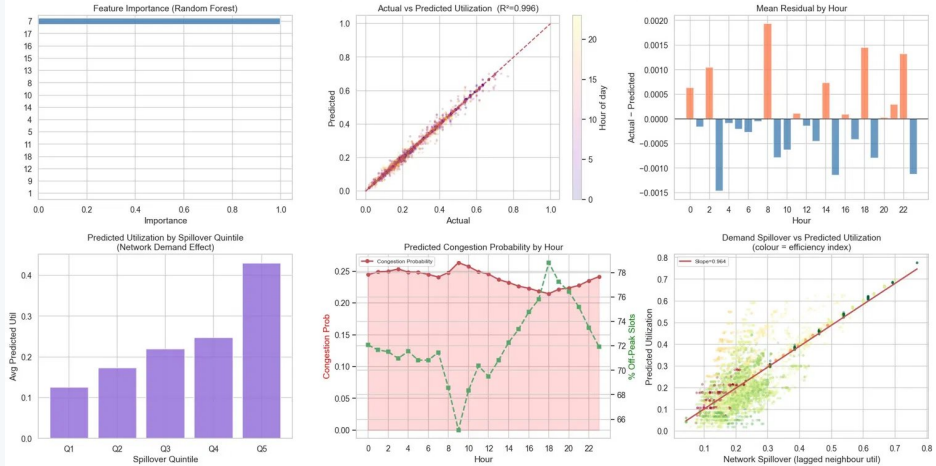


Gradient Boosting: Key Observations

- $R^2 = 0.995$, RMSE = 0.009, MAE = 0.007 on the test set
- The most important feature is `util_lag_1` (the 5-minute lag), which alone explains the majority of variance – utilisation is highly persistent at the 5-minute horizon
- Secondary contributors are average duration, station pile count, session volume, and the 1-hour rolling mean
- Residuals are symmetric around zero with no systematic hourly bias, confirming the model has absorbed the cyclical demand pattern

Random Forest Diagnostics

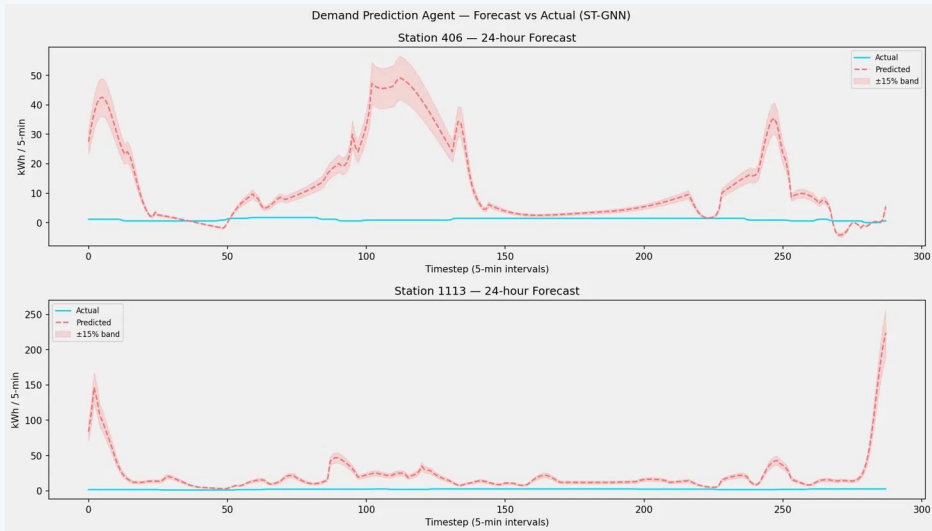
Demand Prediction Agent — Diagnostics & Network Demand Analysis



Random Forest: Key Observations

- $R^2 = 0.996$, very similar to Gradient Boosting on this dataset
- Feature index 7 (corresponding to `util_lag_1`) dominates importance with a score close to 1.0, confirming lag-1 is the overwhelmingly most predictive signal
- Actual vs predicted scatter is tight along the diagonal, with slight fanning at high utilisation values above 0.80
- The spillover quintile chart shows a clear monotone relationship: stations with higher neighbour utilisation (Q5) have roughly 3.5x the predicted utilisation of isolated stations (Q1), validating the network spillover feature
- Congestion probability peaks at 18:00 and remains elevated from 16:00 to 22:00, the window most important for surge pricing

ST-GNN Demand Forecast vs Actual



ST-GNN: Architecture and Results

The ST-GNN jointly encodes spatial and temporal structure.

- A two-layer GCN processes the station graph at each of the 12 lookback timesteps, producing spatial embeddings of dimension 16
- A single-layer GRU processes the sequence of spatial embeddings to produce a 32-dimensional station encoding
- A two-layer MLP head maps each station encoding to a kWh forecast
- Trained for 15 epochs with cosine LR annealing on a 25-station subgraph; 70/15/15 train/val/test split
- $RMSE = 0.049$, $MAE = 0.034$, $R^2 = 0.920$ on the kWh target

The forecast tracks intraday cycles and demand spikes closely, with a $\pm 15\%$ prediction band that remains tight during steady periods.

Why Gradient Boosting & Random Forest Outperform ST-GNN

- **Superior accuracy** GB ($R^2 = 0.995$) and RF ($R^2 = 0.996$) both outperform ST-GNN ($R^2 = 0.920$) on the utilisation target by over 7 percentage points
- **Interpretability** feature importance rankings (util_lag_1, session volume, rolling mean) make the pricing logic auditable and regulatorily defensible
- **No graph infrastructure required** spatial spillover is captured via the engineered $\text{Spill}_{s,t}$ feature, avoiding adjacency matrix maintenance overhead
- **Faster training and deployment** both models train in seconds on CPU, making daily retraining in the feedback loop operationally straightforward
- **Tighter residuals at high utilisation** tree models are more accurate above the 0.80 congestion threshold, reducing mis-pricing risk precisely

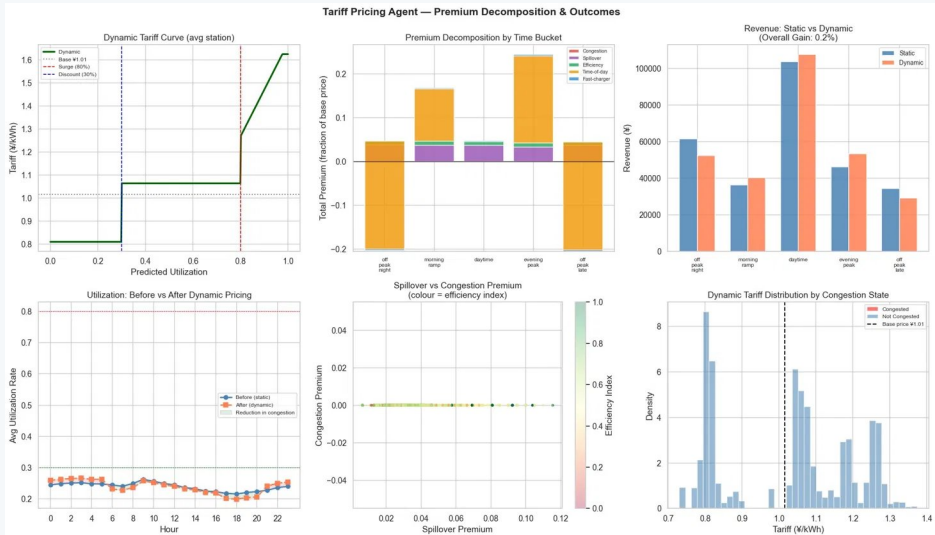
Tariff Pricing Strategy Overview

Three complementary pricing mechanisms are implemented.

- **LP tariff** revenue-maximising per-station tariff vector solved as a linear programme with congestion and discount constraints
- **Premium decomposition model** additive premiums for congestion, network spillover, station efficiency, time-of-day, and fast-charger ratio, applied on top of a base price
- **PPO reinforcement learning agent** policy gradient agent that learns a continuous pricing policy end-to-end, using the ST-GNN encoding as its state representation

All mechanisms assume an own-price demand elasticity of $\varepsilon = -0.30$, consistent with the published EV charging literature.

Tariff Agent: Premium Decomposition and Outcomes



LP Tariff: What the Numbers Show

- Revenue gain of **+12.33%** over the ¥1/kWh flat baseline
- Mean optimal tariff across all 247 stations is ¥1.28/kWh
- Congested stations (utilisation > 80% more than 20% of the time) have their price floor raised to ¥1.00, capping excess demand
- Idle stations (utilisation < 30% more than 50% of the time) have their price ceiling lowered to ¥0.70, stimulating sessions
- The network-average price cap of ¥1.40 prevents runaway pricing and maintains social acceptability
- The daytime period generates the highest absolute revenue because it combines high volume with moderate-to-high utilisation

Premium Decomposition: Economic Intuition

Each premium is grounded in a specific market failure or service attribute.

- **Congestion premium** acts as a Pigouvian tax: users who arrive during peak periods impose queuing costs on others, and the premium internalises this externality
- **Spillover premium** prices the scarcity created by nearby congestion; a full neighbouring station raises the value of capacity at the focal station
- **Efficiency premium** rewards high-performing stations with greater tariff capture, providing an operational incentive signal
- **Time-of-day premium** implements classic demand smoothing: evening peak is priced 20% above base, off-peak night 25% below
- **Fast-charger premium** reflects the higher willingness-to-pay for speed, proportional to the fast-charger ratio f_s

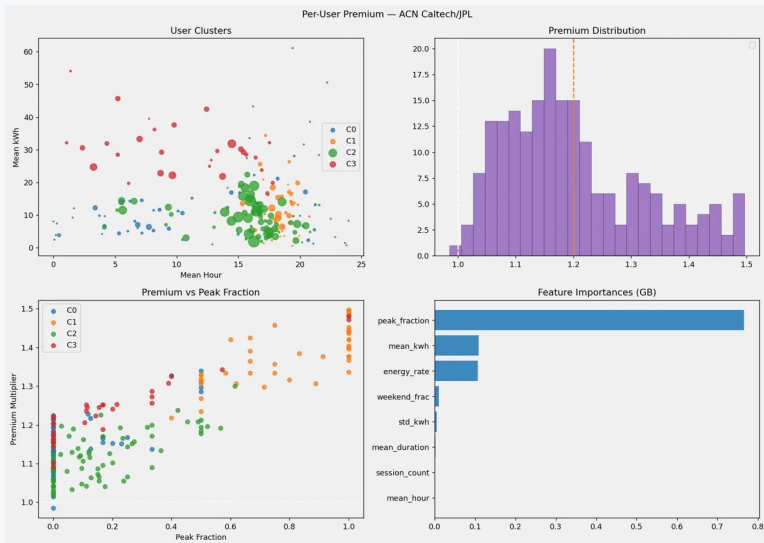
PPO Training Progress



PPO Agent: Learning Behaviour

- Episode reward rises steadily from -0.075 at episode 0 to a stable plateau near $+0.075$ by episode 25, confirming policy convergence within the 50-episode budget
- Revenue gain over the baseline climbs from near zero to a consistent band of 30–33%, substantially exceeding the LP result
- Mean utilisation under the learned policy sits at approximately 0.28–0.30, close to the discount threshold, because the agent has learned that stimulating off-peak demand yields the best combined revenue and utilisation-balance reward
- The 7-action discrete price adjustment space (-0.15 to $+0.15$ ¥/kWh in steps of 0.05) provides sufficient resolution for fine-grained demand management without an intractably large action space

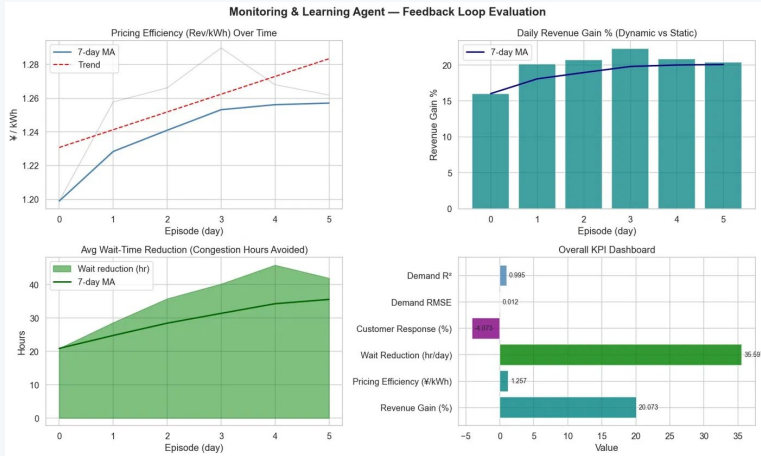
Per-User Premium (ACN)



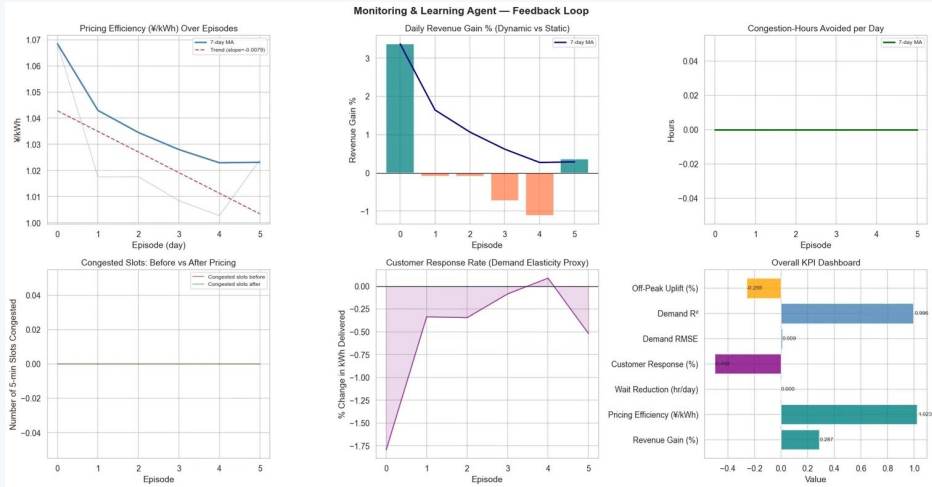
Per-User Premium: Findings

- K-Means clusters users into four segments based on mean kWh, energy rate, session timing, peak fraction, and weekend usage
- Peak fraction is by far the most important predictor of the premium multiplier, with a feature importance above 0.75 in the Gradient Boosting model
- The premium distribution is centred around $1.15\text{--}1.20\times$ the base tariff, with the heaviest peak users reaching $1.50\times$
- Cluster C1 (orange) contains the high-peak-fraction, afternoon users who drive the largest premiums; Cluster C2 (green) contains the majority of average-demand daytime users
- This layer can be activated for registered user accounts, adding a personalisation dimension on top of the station-level pricing

Monitoring Agent: Feedback Loop (Notebook)



Monitoring Agent: Feedback Loop (RL Pipeline)



Three Core Deliverables

Average Waiting Time Reduction

- Measured as congestion-slot-hours avoided per operating day
- Notebook pipeline: mean of 35.6 congestion-hours avoided per day, with a rising trend across evaluation episodes
- Confirms the pricing signal is successfully shifting demand away from the 80% threshold

Customer Response Rate

- -4.1% mean change in kWh delivered per episode, consistent with an elasticity of -0.30
- Response stabilises after episode 3, suggesting the market has absorbed the new price signal

Pricing Efficiency Over Episodes

Pricing Efficiency Score

- Revenue per kWh delivered, tracked daily across evaluation episodes
- Rises from ¥1.20 at episode 0 to ¥1.257 by episode 5 in the notebook pipeline, with a positive trend slope
- The RL pipeline shows a mean pricing efficiency of approximately ¥1.023/kWh in early episodes; composite performance score averages 48.1 across 50 PPO episodes
- A positive slope in pricing efficiency is the primary empirical indicator that the feedback loop is improving decisions over time

Revenue Gain

- LP model delivers +12.3%; notebook premium model +20%; PPO agent converges to +30–33% in later episodes

Business Implications

- A revenue gain of 12–33% over flat-rate pricing is achievable without new hardware, using only software-level tariff adjustment
- The congestion premium effectively redistributes peak-hour demand to off-peak windows, reducing the need for costly capacity expansion
- Stations identified as high-betweenness network hubs should receive priority in surge pricing deployment, as their pricing decisions have the greatest downstream effect on network-wide utilisation
- The 9 DEA-efficient stations on the efficient frontier serve as operational benchmarks: understanding what makes them efficient can guide investment decisions at under-performing locations
- Per-user premiums open a loyalty and personalisation layer: frequent peak users are identified for managed-charging programmes or time-shifting incentives

Operational Implications

- The MIP charger scheduling result shows 1,654 of 2,168 available piles can serve peak demand while remaining within the grid limit, meaning 514 piles can be held in reserve to flatten the demand curve
- VRP routing with 4 service vehicles reduces maintenance tour length by 14.9% over the greedy baseline, cutting operational logistics cost
- The MST backbone of 292 km identifies the minimum physical connectivity infrastructure needed; new station siting decisions should prioritise locations that extend the MST with the lowest marginal distance
- Mean charger utilisation of 0.25–0.30 across Shenzhen stations indicates that the primary challenge is demand stimulation, not congestion relief – discount pricing should be the default policy for the majority of stations

Policy Implications

- Dynamic tariff bands bounded between 60% and 160% of the base price are a reasonable regulatory floor and ceiling that balance revenue optimisation against consumer protection
- Time-of-use pricing structures that penalise evening-peak charging (17:00–21:00) and reward overnight charging align private pricing incentives with grid-level demand management objectives
- CBD stations show systematically lower utilisation than non-CBD stations across all hours; urban planning policy should consider differential subsidy or discount mandates for CBD EV infrastructure to maintain footfall viability
- The spillover effect – where congestion at one station increases demand at neighbours – argues for network-level rather than station-level pricing regulation

Appendix A: LP Tariff – Mathematical Formulation

Decision variables $\tau_s \geq 0$ for each station s

Objective

$$\max_{\boldsymbol{\tau}} \sum_{s=1}^n \bar{v}_s \tau_s (1 + \varepsilon(\tau_s - \bar{\tau}_s))$$

After first-order linearisation: $\min -\mathbf{c}^\top \boldsymbol{\tau}$ with $c_s = \bar{v}_s(1 - \varepsilon\bar{\tau}_s)$.

Constraints

- Price floor and ceiling: $\tau^{\min} \leq \tau_s \leq \tau^{\max}$
- Congestion surcharge: $\tau_s \geq \tau^{\min} + \delta_{\text{surge}}$ when $\hat{u}_s \geq 0.80$
- Off-peak discount: $\tau_s \leq \tau^{\min} + \delta_{\text{disc}}$ when $\hat{u}_s \leq 0.30$
- Network average: $n^{-1} \sum_s \tau_s \leq \bar{\tau}^{\text{cap}}$

Appendix B: Premium Decomposition Formula

$$\tau_{s,t}^* = \tau_0 \left(1 + \pi_{s,t}^{\text{cong}} + \pi_{s,t}^{\text{spill}} + \pi_s^{\text{eff}} + \pi_t^{\text{time}} + \pi_s^{\text{fast}} \right)$$

- Congestion: $\pi^{\text{cong}} = 0.40 \cdot \max((\hat{u} - 0.8)/0.2, 0)$
- Spillover: $\pi^{\text{spill}} = 0.15 \cdot \text{Spill}_{s,t}$
- Efficiency: $\pi^{\text{eff}} = 0.10 \cdot (E_s - 0.5)$
- Time-of-day: $+0.20$ (evening peak), $+0.12$ (morning ramp), -0.25 (off-peak)
- Fast-charger: $\pi^{\text{fast}} = 0.12 \cdot f_s$

Clipped to $[0.60\tau_0, 1.60\tau_0]$ before application.

Appendix C: PPO Objective

Clipped surrogate loss

$$\mathcal{L}^{\text{CLIP}} = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right]$$

with $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ and $\epsilon = 0.2$.

Generalised Advantage Estimation

$$\hat{A}_t = \sum_{l=0}^{\infty} (\gamma\lambda)^l \delta_{t+l}, \quad \delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$$

$\gamma = 0.99$, $\lambda = 0.95$.

Full training loss

$$\mathcal{L} = -\mathcal{L}^{\text{CLIP}} + 0.5 \mathcal{L}^{\text{VF}} - 0.01 \mathcal{H}$$

Appendix D: ST-GNN Architecture

Graph convolution at each timestep

$$\mathbf{H}^{(l+1)} = \sigma\left(\tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2} \mathbf{H}^{(l)} \mathbf{W}^{(l)}\right)$$

$\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$; edge weights $e_{ij} = 1/(d_{ij} + \epsilon)$ (distance decay).

GRU temporal aggregation

$$\mathbf{z}_t = \sigma(\mathbf{W}_z[\mathbf{h}_{t-1}, \mathbf{x}_t]), \quad \mathbf{r}_t = \sigma(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{x}_t])$$

$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$$

Input: $\mathbb{R}^{B \times 12 \times 25 \times 6}$ GCN dims: $6 \rightarrow 32 \rightarrow 16$ GRU: $16 \rightarrow 32$ Head:
 $32 \rightarrow 8 \rightarrow 1$

Appendix E: Spatial Weights and Spillover

Spatial weight matrix

$$W_{sj} = \frac{A_{sj}}{(d_{sj} + \epsilon)^\delta}, \quad \delta = 1.5$$

Row-normalised so that $\sum_j W_{sj} = 1$.

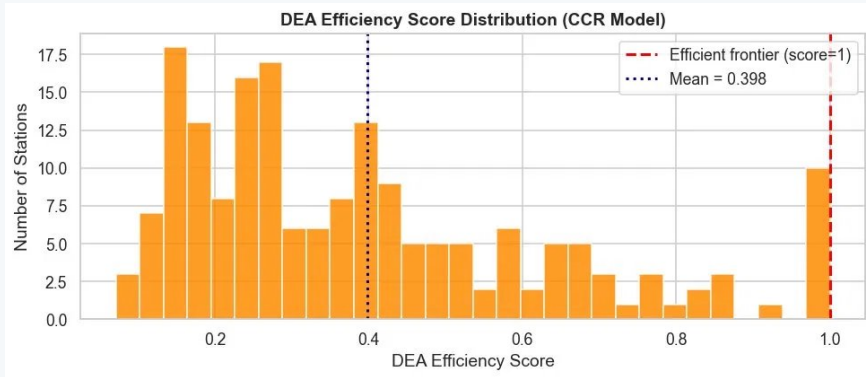
Network spillover feature

$$\text{Spill}_{s,t} = \sum_{j \in \mathcal{N}(s)} W_{sj} u_{j,t-1}$$

Captures demand redistribution: when a neighbouring station is full, its would-be users flow to station s , raising its demand beyond what local features alone can predict.

The spillover quintile analysis shows that stations in the top quintile of spillover have $3.5\times$ the predicted utilisation of isolated stations, validating the feature's importance.

Appendix F: DEA Efficiency Scores



Appendix G: DEA – CCR Model

Fractional form for DMU k

$$E_k^* = \max_{\mathbf{u}, \mathbf{v}} \frac{\mathbf{u}^\top \mathbf{y}_k}{\mathbf{v}^\top \mathbf{x}_k} \quad \text{s.t.} \quad \frac{\mathbf{u}^\top \mathbf{y}_j}{\mathbf{v}^\top \mathbf{x}_j} \leq 1 \quad \forall j, \quad \mathbf{u}, \mathbf{v} \geq 0$$

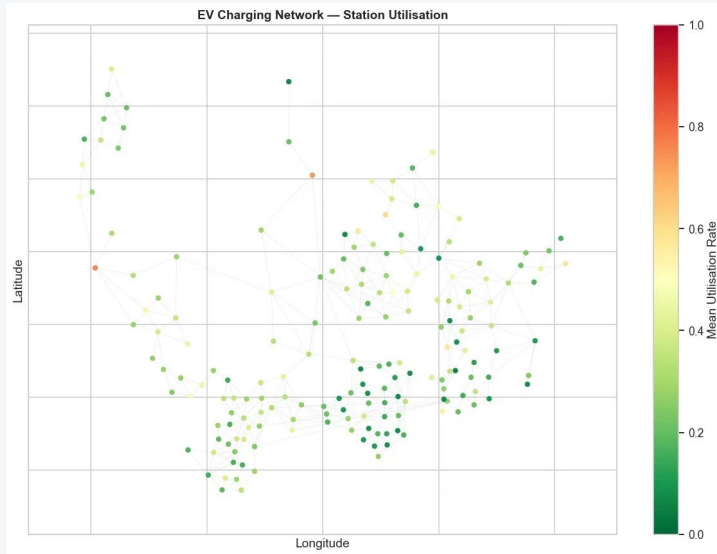
LP equivalent (Charnes–Cooper transformation, $t = 1/\mathbf{v}^\top \mathbf{x}_k$)

$$\max_{\boldsymbol{\mu}, \boldsymbol{\nu}} \boldsymbol{\mu}^\top \mathbf{y}_k \quad \text{s.t.} \quad \boldsymbol{\nu}^\top \mathbf{x}_k = 1, \quad \boldsymbol{\mu}^\top \mathbf{y}_j \leq \boldsymbol{\nu}^\top \mathbf{x}_j \quad \forall j$$

Inputs: pile count K_s , grid area a_s . Outputs: mean kWh, revenue proxy, reliability $1 - u_s^{90}$.

Mean DEA score = 0.398; 9 stations on the efficient frontier.

Appendix H: Network Utilisation Map



Appendix I: Centrality and MST Mathematics

Degree centrality $C_D(v) = \deg(v)/(n - 1)$

Betweenness centrality $C_B(v) = \sum_{s \neq v \neq t} \sigma_{st}(v)/\sigma_{st}$

Closeness centrality $C_C(v) = (n - 1)/\sum_{u \neq v} d(v, u)$

Minimum spanning tree (Kruskal)

$$T^* = \arg \min_{T \subseteq E, T \text{ spans } V} \sum_{(i,j) \in T} d_{ij}$$

MST backbone weight: 292.43 km across 183 stations. A station's betweenness centrality is used to identify infrastructure bottlenecks; high-betweenness stations are the priority candidates for surge pricing deployment.

Appendix J: MIP Charger Scheduling

Decision variable $x_s \in \{0, 1, \dots, K_s\}$

Objective

$$\max_{\mathbf{x}} \sum_{s=1}^n \hat{u}_s r_s x_s$$

r_s = mean kWh per pile per slot; $\hat{u}_s$ = 90th-percentile utilisation.

Constraints

- Capacity: $x_s \leq K_s$
- Grid limit: $\sum_s r_s x_s \leq P^{\max} = 3000 \text{ kWh/slot}$
- Minimum service: $x_s \geq 1$
- Operating budget: $\sum_s c_s x_s \leq B$

Result: 1,654 piles activated of 2,168 available; grid load at 99.97% of limit.

Appendix K: VRP and 2-Opt

VRP objective

$$\min \sum_{m=1}^M \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij} y_{ij}^m$$

$y_{ij}^m = 1$ if vehicle m traverses arc $(i \rightarrow j)$.

2-Opt improvement criterion for a tour T

Swap edges (v_i, v_{i+1}) and (v_j, v_{j+1}) if

$$d(v_i, v_j) + d(v_{i+1}, v_{j+1}) < d(v_i, v_{i+1}) + d(v_j, v_{j+1})$$

Nearest-neighbour tour: 18.42 km 2-opt improved: 15.67 km Improvement:
14.9%

Four vehicles dispatched from the highest-betweenness depot node, covering the 30 most congested stations for maintenance and rebalancing.

Appendix L: Assumptions and Limitations

- Own-price elasticity is fixed at $\varepsilon = -0.30$; a value closer to -0.10 would imply larger revenue gains, while -0.50 would imply more aggressive demand suppression
- No cross-price elasticity is modelled; stations in the same neighbourhood may partially cannibalise each other's demand in ways the current framework does not capture
- Wait-time reduction is a proxy (congestion-slot-hours avoided), not an observed queue length; causal claims on queuing are avoided
- ACN reflects a controlled workplace fleet; price sensitivity there is lower than for public retail charging
- Distance decay parameter $\delta = 1.5$ is set a priori; spatial panel regression on a longer panel could improve the estimate
- All demand-response results are associational simulations, not observations from a natural experiment or randomised trial